

3.6 Directional Derivatives and Gradients_Guide

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3.6
Direction...

3.6 Directional Derivatives and Gradients

Recall: Suppose we have a surface given by the two variable function $z = f(x, y)$.

If the coordinates of P are (x_0, y_0, z_0) , then the partial derivative $f_x(x_0, y_0)$ is the slope of the line tangent to the curve C_1 at P ,

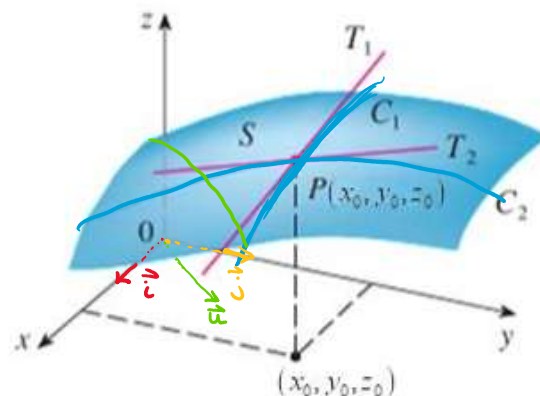
or we can say that it is the rate of change of z at P in the

direction of of a unit vector $\vec{i} = \langle 1, 0 \rangle$

Moreover, the partial derivative $f_y(x_0, y_0)$ is the slope of

the line tangent to the curve C_2 at P , or we can say that it is the rate of change of z at P in the direction of

a unit vector $\vec{j} = \langle 0, 1 \rangle$



Motivation: Suppose we wanted to move across $z = f(x, y)$ in a direction that was not parallel to $\mathbf{i}$ or $\mathbf{j}$. How could we determine the rate of change of z at P in the direction of an arbitrary unit vector $\mathbf{u} = \langle a, b \rangle$?

Definition: The directional derivative of f at (x_0, y_0) in the direction of a unit vector $\mathbf{u} = \langle a, b \rangle$ is

$$D_{\mathbf{u}}f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}.$$

Note: If $\mathbf{u} = \langle 1, 0 \rangle$ we get the definition of the first x partial, and if $\mathbf{u} = \langle 0, 1 \rangle$ we get the definition of the first y partial!

Note: It is extremely important to remember that directional derivatives are always computed in the direction of a UNIT VECTOR!!!!!!

The next theorem gives us an easy way to compute directional derivatives. As long as our starting function is differentiable.

Theorem: Suppose that f is a differentiable function of x and y , then f has a directional derivative in the direction of any unit vector $\mathbf{u} = \langle a, b \rangle$ and $D_{\mathbf{u}}f(x, y) = f_x(x, y)a + f_y(x, y)b$.

Proof: Suppose g is a function of a single variable $g(h) = f(x_0 + ha, y_0 + hb)$.

$$\text{So } g'(0) = \lim_{h \rightarrow 0} \frac{g(0+h) - g(0)}{h} = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h} = D_{\mathbf{u}}f(x_0, y_0)$$

$g = f$ Note if $x = x_0 + ha$ & $y = y_0 + hb$ then $g(h) = f(x, y)$ & by the Chain Rule

$$g'(h) = \frac{\partial f}{\partial x} \cdot \frac{dx}{dh} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dh} = f_x(x, y) \cdot a + f_y(x, y) \cdot b$$

$\begin{matrix} g=f \\ \swarrow \searrow \\ x \quad y \\ | \quad | \\ h \quad h \end{matrix}$ So $g'(0) = f_x(x_0, y_0) \cdot a + f_y(x_0, y_0) \cdot b$

$$\text{Thus } D_{\mathbf{u}}f(x_0, y_0) = f_x(x_0, y_0) \cdot a + f_y(x_0, y_0) \cdot b. \quad \blacksquare$$

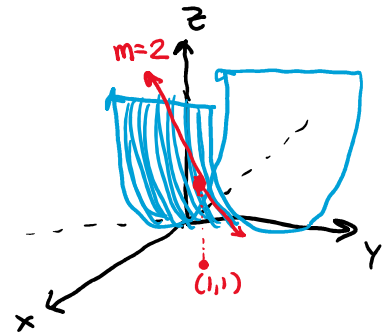
Example 1: Consider the function $f(x, y) = x^2$.

- a) Find the directional derivative of the function $f(x, y) = x^2$ at the point $(1, 1)$ in the direction of the vector $\mathbf{v} = \langle 1, 0 \rangle$.

Note $\mathbf{v}$ is already a unit vector.

$$D_{\mathbf{v}}f(x, y) = (2x) \cdot 1 + (0) \cdot 0 = 2x$$

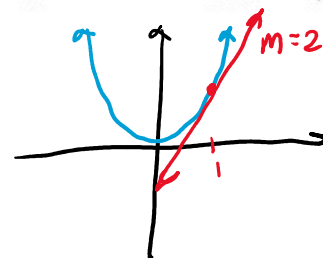
$$\text{So } D_{\mathbf{v}}f(1, 1) = 2(1) = 2$$



- b) How does your answer in part (a) relate to the slope of the tangent line on the graph of $f(x) = x^2$ at the point when $x = 1$?

$$f'(x) = 2x$$

$$f'(1) = 2$$



Example 2: Find the directional derivative of the function $f(x, y) = \frac{x}{x^2 + y^2}$ at the point (1,2) in the direction of the vector $\mathbf{v} = \langle 3, 5 \rangle$. (Hint: Directional derivatives must be computed in the direction of a unit vector!)

$$\text{let } \vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{\sqrt{34}} \langle 3, 5 \rangle = \left\langle \frac{3}{\sqrt{34}}, \frac{5}{\sqrt{34}} \right\rangle$$

$$f_x = \frac{(x^2 + y^2)(1) - x(2x)}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$f_y = \frac{(x^2 + y^2)(0) - x(2y)}{(x^2 + y^2)^2} = \frac{-2xy}{(x^2 + y^2)^2}$$

$$D_{\vec{u}} f(x, y) = f_x \cdot a + f_y \cdot b$$

$$D_{\vec{u}} f(1, 2) = f_x(1, 2) \frac{3}{\sqrt{34}} + f_y(1, 2) \frac{5}{\sqrt{34}}$$

$$= \frac{3 \cdot 3}{25\sqrt{34}} + \frac{-4 \cdot 5}{25\sqrt{34}}$$

$$= \frac{-11}{25\sqrt{34}}$$

Question to the Class:

Recall that if f is a differentiable function of two variables and $\mathbf{u} = \langle a, b \rangle$ is a unit vector, then

$D_{\mathbf{u}} f(x, y) = f_x(x, y)a + f_y(x, y)b$. Can we rewrite this formula as a dot product as vectors?

Yes!

$$D_{\vec{u}} f(x, y) = \underbrace{\langle f_x(x, y), f_y(x, y) \rangle}_{\text{I'd like to name this vector}} \cdot \underbrace{\langle a, b \rangle}_{\vec{u}!}$$

Definition: If f is a function of two variables x and y , then the gradient of f is the vector function $\vec{\nabla}f$ defined by $\vec{\nabla}f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j}$.

Note: With this definition we have that $D_{\mathbf{u}}f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle \cdot \mathbf{u} = \vec{\nabla}f(x, y) \cdot \mathbf{u}$.

Example 3: Find the directional derivative of the function $f(x, y) = x^2y^3 - 4y$ at the point $(2, -1)$ in the direction of the vector $\mathbf{v} = 2\mathbf{i} + 5\mathbf{j}$.

$$\|\mathbf{v}\| = \sqrt{4 + 25} = \sqrt{29}. \text{ So let } \vec{u} = \frac{1}{\sqrt{29}} \langle 2, 5 \rangle = \left\langle \frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}} \right\rangle$$

$$\text{Also } \vec{\nabla}f = \left\langle \underbrace{2xy^3}_{f_x}, \underbrace{3x^2y^2 - 4}_{f_y} \right\rangle$$

$$D_{\vec{u}}f(x, y) = \vec{\nabla}f \cdot \vec{u} = 2xy^3 \cdot \frac{2}{\sqrt{29}} + (3x^2y^2 - 4) \frac{5}{\sqrt{29}}$$

$$D_{\vec{u}}f(2, -1) = \frac{-8}{\sqrt{29}} + \frac{40}{\sqrt{29}} = \frac{32}{\sqrt{29}}$$

The notion of directional derivative extends in a natural way to functions of more than two variables. The next few definitions address this.

Definition: Suppose that f is a function such that $f: \mathbb{R}^n \rightarrow \mathbb{R}$ and $\mathbf{u}$ is a unit vector in V_n . Then, the directional derivative of f at $\mathbf{x}_0 \in \mathbb{R}^n$ in the direction of $\mathbf{u}$ is $D_{\mathbf{u}}f(\mathbf{x}_0) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x}_0 + h\mathbf{u}) - f(\mathbf{x}_0)}{h}$.

Definition: If f is a function of three variables, then the gradient vector of f is $\vec{\nabla}f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k} = \langle f_x, f_y, f_z \rangle$.

Theorem: Just as with functions of two variables, if f is a differentiable function of three variables and $\mathbf{u}$ is a unit vector, then $D_{\mathbf{u}}f(x, y, z) = \vec{\nabla}f(x, y, z) \cdot \mathbf{u}$.

Proof: One only needs to extend the proof for two variable functions.

Note: The notion of gradient and the above theorem extends beyond three variable functions.

Example 4: Suppose that $f(x, y, z) = xyz$.

a) Find the gradient of f .

$$\vec{\nabla} f = \langle yz, xz, xy \rangle$$

b) Find the directional derivative of f at $(1, 3, 0)$ in the direction of $\langle 1, 2, 2 \rangle$.

$$\text{let } \vec{u} = \frac{1}{3} \langle 1, 2, 2 \rangle = \langle \frac{1}{3}, \frac{2}{3}, \frac{2}{3} \rangle$$

$$\begin{aligned} D_{\vec{u}} f(1, 3, 0) &= \vec{\nabla} f(1, 3, 0) \cdot \vec{u} \\ &= \langle 0, 0, 3 \rangle \cdot \langle \frac{1}{3}, \frac{2}{3}, \frac{2}{3} \rangle = 2 \end{aligned}$$

c) How should the unit vector $\vec{u}$ be chosen so that $D_{\vec{u}} f(1, 3, 0)$ is maximized?

$$\text{We would select } \vec{u} = \langle 0, 0, 1 \rangle$$

$$\& D_{\vec{u}} f(1, 3, 0) = 3.$$

Exploration:

Suppose that f is a two or three variable function which is differentiable. Suppose that we want to find a unit vector $\vec{u}$ so that $D_{\vec{u}} f(\vec{x})$ is as large as possible.

$$D_{\vec{u}} f(\vec{x}) = \vec{\nabla} f(\vec{x}) \cdot \vec{u} = \|\vec{\nabla} f(\vec{x})\| \cdot \|\vec{u}\| \cos(\theta) = \|\vec{\nabla} f(\vec{x})\| \cos(\theta)$$

This is maximize when $\cos(\theta) = 1$ if $\theta = 0$ or 2π

This implies that $\vec{u}$ & $\vec{\nabla} f(\vec{x})$ are parallel & point the same direction!

$$\text{So } \vec{u} = \frac{\vec{\nabla} f(\vec{x})}{\|\vec{\nabla} f(\vec{x})\|}.$$

Question: What should we take $\vec{u}$ to be to make the directional derivative as small as possible?

Theorem: Suppose that f is a differentiable function of two or three variables.

The maximum value of the directional derivative $D_{\vec{u}} f(\vec{x})$ is $\|\vec{\nabla} f(\vec{x})\|$ and it occurs when $\vec{u}$ has the same direction as the gradient vector $\vec{\nabla} f(\vec{x})$.

Moreover, the minimum value of the directional derivative $D_{\vec{u}} f(\vec{x})$ is $-\|\vec{\nabla} f(\vec{x})\|$ and it occurs when $\vec{u}$ points in the opposite direction of the gradient vector $\vec{\nabla} f(\vec{x})$.

Note: We will often sum up the above theorem by saying that the gradient of a function points in the direction of the function's steepest increase.

Example 5: Suppose you are skiing on a mountain whose surface is given by the equation $z = -x^2 - y^2 + 6$. You are currently at the point $(1, 2, 1)$ on the surface.

- a) In the direction of which unit vector should you ski if you want to ski in the direction of the steepest decrease?

$$\text{We see } \vec{\nabla} f = \langle -2x, -2y \rangle$$

$$\& \vec{\nabla} f(1, 2) = \langle -2, -4 \rangle \leftarrow \text{direction of biggest uphill}$$

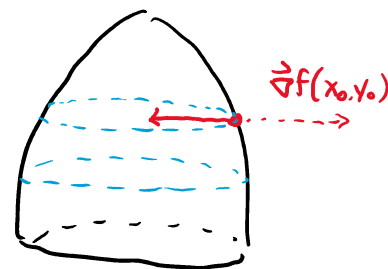
$$\text{So we need } -\langle -2, -4 \rangle = \langle 2, 4 \rangle \leftarrow \text{direction of biggest downhill.}$$

$$\text{So let } \vec{u} = \left\langle \frac{2}{\sqrt{20}}, \frac{4}{\sqrt{20}} \right\rangle$$



- b) What is the rate of decrease when you travel in the direction of the unit vector you found in (a)?

$$-\|\vec{\nabla} f(1, 2)\| = -\sqrt{20}$$



Note: We can think of the gradient $\vec{\nabla} f(x, y)$ as a “compass needle” that points us in the direction of greatest increase at any given point on our surface. It is important to note that this “compass needle” only consists of an x and y direction and doesn’t depend on z in any way.